Accelerated Math II/III Name_____

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HW #6 - Review #1

Date_____ Period____

Write the polynomial in standard form. Name each polynomial by degree and number of terms.

1) $-3p^7$

2) $-4r + 7r^6$

Perform the indicated operation.

3) $h(n) = -2n + 2$
$g(n) = n^2 + 1$
Find $h(n) - g(n)$

4) $h(t) = -3t$
$g(t) = 2t^3 - 5$
Find $(h \cdot g)(t)$

5) $g(a) = a^2 - 5$
$h(a) = 3a - 3$
Find $g(a) \div h(a)$

6) $g(a) = -4a + 3$
$h(a) = a^2 + 4$
Find $g(-3) \cdot h(-3)$

7) $g(n) = 3n + 5$
$f(n) = n^2 + 5$
Find $\left(\frac{g}{f}\right)(1)$

8) $g(a) = 4a + 3$
$f(a) = 3a^3 - a^2$
Find $g(-4t) - f(-4t)$

9) $h(a) = 2a$
$g(a) = 2a + 2$
Find $h(a - 3) \cdot g(a - 3)$

10) $f(x) = 3x - 3$
$g(x) = x^2 + 3$
Find $\left(\frac{f}{g}\right)\left(\frac{x}{2}\right)$

11) $g(x) = 3x - 3$
$h(x) = x^3 - 2$
Find $g(h(x))$

12) $f(x) = x - 1$
Find $f(f(1))$

13) $g(x) = x^3 + 2x$
$h(x) = 2x + 5$
Find $g(h(-2))$

14) $f(x) = 4x - 4$
Find $(f \circ f)(3x)$

15) $g(a) = a - 2$
$h(a) = -2a^2 - 2a$
Find $g\left(h\left(\frac{a}{3}\right)\right)$

16) $g(n) = n^2 + 2$
$f(n) = 2n + 5$
Find $(5g - 4f)(n)$

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17) $g(n) = 2n + 5$
$h(n) = n^3 - 2$
Find $(4g + 3h)(-3)$

18) $g(n) = n^2 - 4$
$h(n) = 3n$
Find $2g\left(\frac{n}{3}\right) + 3h\left(\frac{n}{3}\right)$

Find the zeros of the function. Then graph the function using your graphing calculator.

19) $f(x) = x(x - 4)(x + 2)$

20) $f(x) = x(3x + 5)(3x - 4)$

Write a polynomial function of least degree with integer coefficients that has the given zeros.

21) $1, \ -\frac{2}{5}, \ \frac{5}{2}$

22) $-3, \ \frac{4}{3}, \ \frac{5}{3}$

Divide using the most appropriate method for the problem. Write your answer in fraction form.

23) $(x^5 - x^4 - 7x^3 + 9x + 28) \div (x - 2)$

24) $(x^2 + 4x) \div (x + 6)$

25) $(12x^3 - 4x^2 + x + 4) \div (6x - 5)$

26) $(2x^4 + 10x^3 + 25x^2 + 23x + 28) \div (x^2 + 2x + 4)$

Determine if $d(x)$ is a factor of $f(x)$.

27) $f(x) = 5x^5 - 8x^4 - 26x^3 + 21x^2 - 22x + 12$
$d(x) = x - 3$

28) $f(x) = 9x^3 + 9x^2 - 10x - 8$
$d(x) = 3x + 4$

29) $f(x) = 18x^3 + 15x^2 - 13$
$d(x) = 6x - 3$

30) $f(x) = 4x^4 - 10x^3 + 5x - 2$
$d(x) = x - 2$

Evaluate $f(x)$ at k using synthetic division and the remainder theorem.

31) $f(x) = 6x^4 + x^3 - 18x^2 - 15x - 2$
$k = 1$

32) $f(x) = 5x^3 - 29x - 18$
$k = 3$

33) $f(x) = 4x^3 - 10x^2 + 8x$
$k = -1$

34) $f(x) = 5x^4 + 4x^3 + 20x^2$
$k = 1$

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Answers to HW #6 - Review #1

1) seventh degree monomial
2) sixth degree binomial
3) $-n^2 - 2n + 1$
4) $-6t^4 + 15t$
5) $\frac{a^2 - 5}{3a - 3}$
6) 195
7) $\frac{4}{3}$
8) $192t^3 + 16t^2 - 16t + 3$
9) $4a^2 - 20a + 24$
10) $\frac{-12 + 6x}{12 + x^2}$
11) $3x^3 - 9$
12) -1
13) 3
14) $48x - 20$
15) $\frac{-2a^2 - 6a - 18}{9}$
16) $5n^2 - 8n - 10$
17) -91
18) $\frac{27n - 72 + 2n^2}{9}$
19) $\{0, 4, -2\}$
20) $\left\{0, -\frac{5}{3}, \frac{4}{3}\right\}$
21) $f(x) = 10x^3 - 31x^2 + 11x + 10$
22) $f(x) = 9x^3 - 61x + 60$
23) $x^4 + x^3 - 5x^2 - 10x - 11 + \frac{6}{x - 2}$
24) $x - 2 + \frac{12}{x + 6}$
25) $2x^2 + x + 1 + \frac{9}{6x - 5}$
26) $2x^2 + 6x + 5 + \frac{-11x + 8}{x^2 + 2x + 4}$
27) Yes
28) Yes
29) No
30) No
31) -28
32) 30
33) -22
34) 29

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